

Truss Adaptors for Hanging & Supporting Box and Triangle Truss

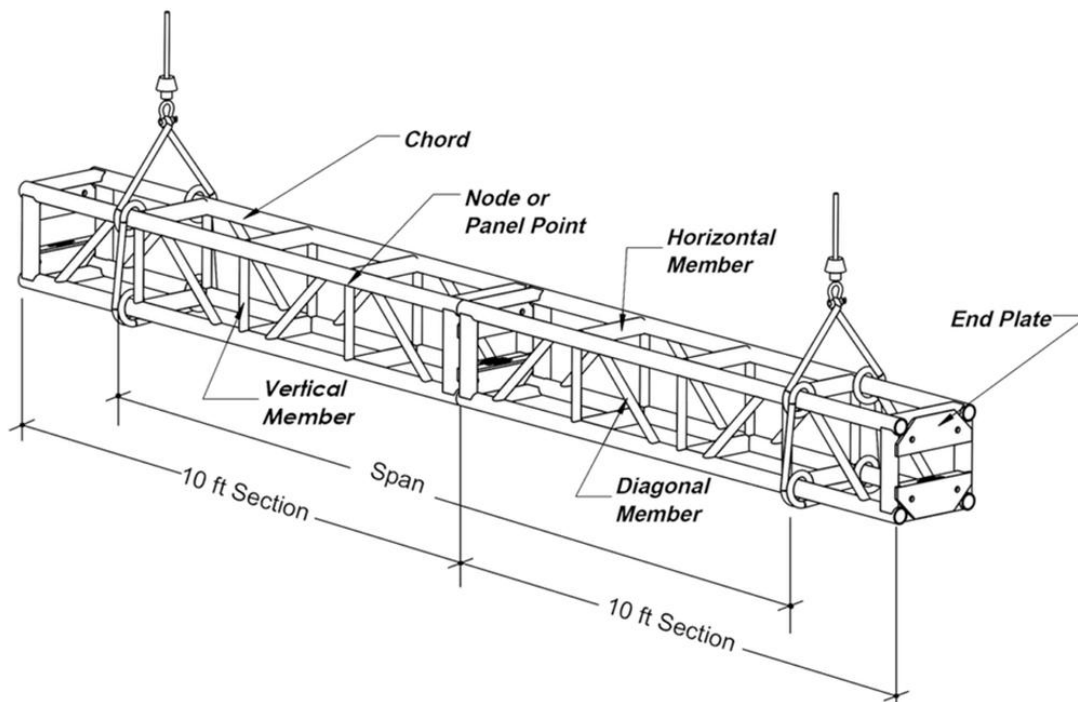
New-starter reference guide: adaptor names, what they look like, how they are normally used, and where they should sit on the truss.

Prepared for Andy Jones / Stage Electrics. This document is a training aid only. Always use the manufacturer manual, rated load data, LOLER inspection status, and a competent rigging assessment for the actual job.

1. Truss component names

Before selecting an adaptor, identify the main chords and avoid confusing them with braces/members. In manufacturer instructions, the words chord, node, panel point, diagonal member and vertical member have specific meaning.

Truss component names



Use these terms when identifying where an adaptor may or may not be attached.

Truss component names. Supplied diagram added for new-starter familiarisation.

2. What a truss adaptor does

A truss adaptor is a rated component used to connect a truss section to a lifting point, winch stand, hoist, or other support. The correct adaptor depends on the truss type, chord diameter, chord centres, intended load direction, WLL/SWL, and the way the truss is being raised or suspended.

Common entertainment truss adaptors are often designed around 48-51 mm truss tubes, but tube diameter alone is not enough. The chord spacing, adaptor adjustment range, WLL/SWL, load direction and truss orientation must all match the manufacturer data.

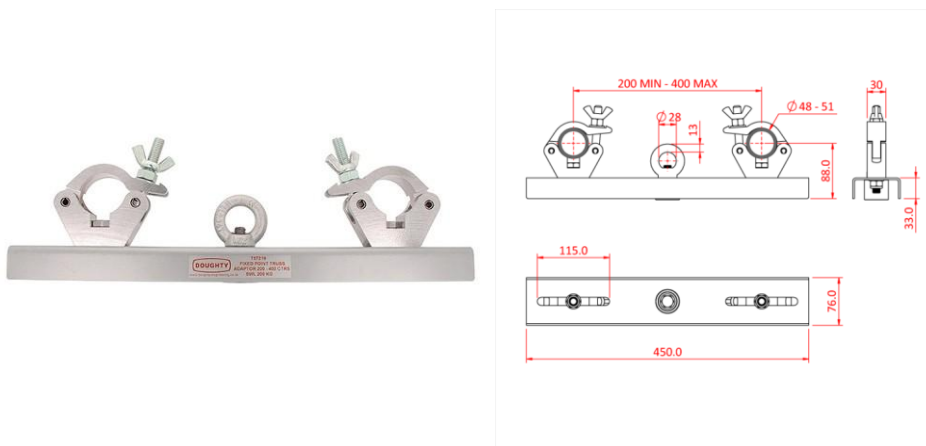
3. Main types of truss adaptors

Adaptor type	Typical use	Box/square truss	Triangle truss	New-starter note
Hanging truss adaptor / suspension adaptor	Suspending truss from a chain hoist or rated lifting point	Yes	Yes	Look for a central eye/ring/bow nut. It is not a stand adaptor.
Solid hanging truss adaptor	Suspending truss; compact or machined style	Yes	Yes	Check the exact WLL and chord centre range. Similar looking items can have different ratings.
Fixed truss adaptor	Mounting truss on winch stands where stands are raised evenly	Yes	Yes	Rigid spigot/support. Do not use it to hang truss from above unless the manufacturer says so.
Swivel truss adaptor	Mounting truss on stands where stands may be raised one at a time	Yes	Yes	The swivel helps during setup. It does not make the stand or truss stronger.
Ladder truss adaptor	Supporting two-chord ladder truss	Usually no	Usually no	For ladder truss, not normally for box or triangle truss.
Fixed point / swivel truss plate	Creating a defined pickup/suspension point	Usually yes	Model-dependent	Used where a planned pickup point is needed. Must match truss geometry and calculations.
System-to-system adaptor	Connecting different truss series/systems	Depends	Depends	Only use if approved by the manufacturer or calculation. Not automatically a hanging point.

4. Adaptor images and descriptions

The diagrams below are deliberately labelled and simplified so that somebody new to the job can identify the main families of adaptor. After this section there are official manufacturer reference links for product photographs, manuals and part-specific data.

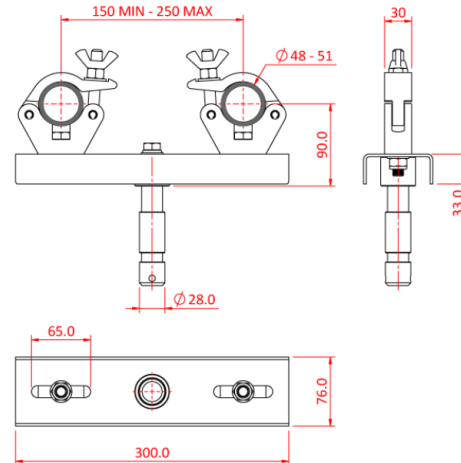
Hanging Truss Adaptor / Suspension Adaptor



Illustrative training diagram - Hanging Truss Adaptor / Suspension Adaptor. Check official manufacturer product image and manual before use.

Used for suspending square/box or triangular truss from a chain hoist or other rated lifting point. The central eye/ring provides the hoist connection. The couplers must sit fully on the main truss chords, not on welds, diagonal members or vertical members.

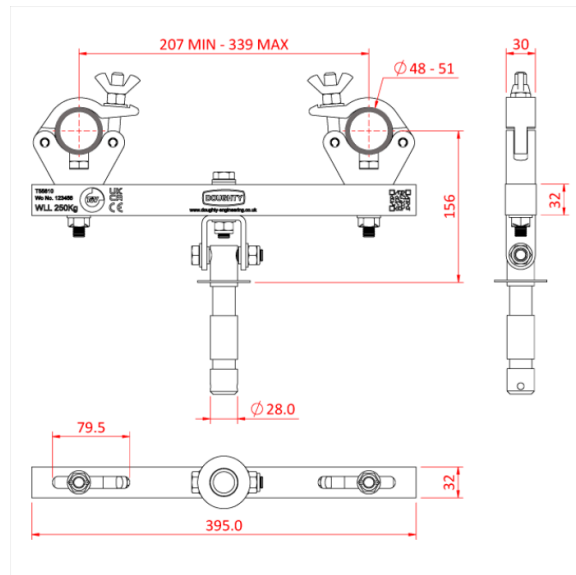
Fixed Truss Adaptor



Illustrative training diagram - Fixed Truss Adaptor. Check official manufacturer product image and manual before use.

Used to support truss on a winch stand or similar stand receiver. The stand supports the adaptor from below via the fixed spigot. This is for support on stands, not for hanging from above.

Swivel Truss Adaptor



Illustrative training diagram - Swivel Truss Adaptor. Check official manufacturer product image and manual before use.

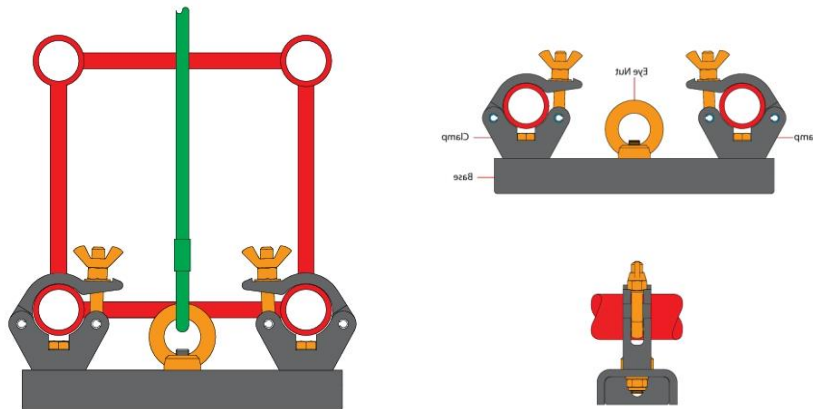
Used for stand-supported truss where one stand may be raised before the other. The swivelling spigot can help reduce binding during setup, but it does not increase the truss or stand rating.

Illustrative training diagram - Solid / Machined Swivel Truss Adaptor. Check official manufacturer product image and manual before use.

A compact machined/solid style of stand adaptor, typically with slimline couplers and a swivelling stand spigot. Check the exact tube centres, WLL, finish, and part number.

5. Do adaptors have to be fitted under the chords?

Fitting position: Under/around main chords



Illustrative orientation principle for a hanging adaptor.

For normal hanging/suspension use, the adaptor body and lifting eye are generally intended to sit below the pair of main chords being picked up, with the couplers fully seated around the chord tubes. The hoist or lifting point then pulls upward on the central eye/ring while the adaptor transfers load through the couplers into the truss chords.

The deciding rule is the manufacturer's intended load direction for that exact adaptor. Do not flip, invert, or use an adaptor in a different orientation simply because the couplers close around the tube. If the manual or data sheet does not show or permit that orientation, do not assume it is acceptable.

For stand adaptors, the adaptor is normally below the truss because the stand/spigot supports the adaptor from underneath and the truss sits in the couplers above the support. For triangular truss, orientation matters more: apex-up and apex-down arrangements change which chord pair is being picked up and how the adaptor sits.

6. New-starter identification checklist

- Read the markings on the adaptor: manufacturer, part number if present, WLL/SWL, and any inspection tag.
- Identify whether it is a hanging adaptor, fixed stand adaptor, swivel stand adaptor, fixed point plate, or system adaptor.
- Check the adaptor is designed for the truss tube diameter and chord spacing.
- Check whether the truss is box/square, triangle, ladder, or another system.
- Check the truss orientation, especially triangle truss apex-up/apex-down.
- Check the couplers sit on main chords only - not on welds, diagonal members, vertical members, or damaged tube.
- Check the intended load direction: hanging from above, supporting from below, or creating a pickup point.
- Escalate to a competent rigger/supervisor if the adaptor does not look exactly like the drawing/manual for the intended use.

7. Do not do this

- Do not clamp onto diagonal bracing or vertical members instead of main chords.
- Do not seat couplers over welds or damaged tube sections.
- Do not leave couplers partly seated, twisted, or skewed on the tube.
- Do not side-load an eye nut, bow nut, hanging ring, or hook unless the manufacturer explicitly allows it.
- Do not use a stand adaptor as a hanging adaptor, or a hanging adaptor as a stand adaptor, unless the manufacturer specifically states that it is designed for that use.
- Do not use a normal half coupler to suspend the truss itself unless that exact arrangement is manufacturer-approved and rated for the purpose.

- Do not mix truss systems or adaptors from different manufacturers in a load-bearing span unless compatibility is approved or engineered.

8. Quick selection guide

Situation	Likely adaptor family
Hanging box/square truss from a hoist	Rated hanging truss adaptor or solid hanging truss adaptor matched to chord diameter, centres and WLL/SWL.
Hanging triangle truss	Rated hanging truss adaptor confirmed for triangular truss; check apex orientation and chord pair.
Supporting truss on two winch stands	Fixed truss adaptors if stands are raised evenly; swivel truss adaptors if the lift sequence needs articulation.
Creating a specific pickup point	Manufacturer-approved fixed point plate or swivel truss plate.
Connecting different truss systems	Only manufacturer-approved system adaptor; never assume mixed systems are structurally compatible.

9. Safety and inspection notes

- Use only adaptors with a visible and suitable WLL/SWL and traceable manufacturer identification.
- Check the product is suitable for the truss tube diameter and chord centres. Tube diameter alone is not enough.
- All imposed loads must be considered, including dynamic effects from lifting machinery and movement during setup.
- Inspect before each use. Remove from service if there is deformation, cracking, corrosion, missing parts, damaged threads, or unclear markings.
- Tighten to the manufacturer instruction. Over-tightening can damage the coupler and/or truss chord.
- For UK lifting operations, treat relevant adaptors as lifting accessories under LOLER arrangements, including thorough examination by a competent person at the required interval or under a written examination scheme.

10. Source notes and useful references

- Doughty Engineering truss adaptors product range: <https://doughty-engineering.co.uk/products/rigging/truss-adaptors/>
- Doughty hanging truss adaptor product page: <https://doughty-engineering.co.uk/products/hanging-truss-adaptor/>
- Doughty fixed truss adaptor product page: <https://doughty-engineering.co.uk/products/fixed-truss-adaptor/>
- Doughty solid swivel truss adaptor product page: <https://doughty-engineering.co.uk/products/solid-swivel-truss-adaptors/>
- Doughty hanging truss adaptors user manual: <https://doughty-engineering.co.uk/wp-content/uploads/2025/03/Hanging-Truss-Adaptors-User-Manual.pdf>
- Doughty fixed truss adaptor user manual: <https://doughty-engineering.co.uk/wp-content/uploads/2025/03/Fixed-Truss-Adaptor-User-Manual.pdf>
- Doughty swivel truss adaptor user manual: <https://doughty-engineering.co.uk/wp-content/uploads/2025/03/Swivel-Truss-Adaptor-User-Manual.pdf>
- HSE guidance: thorough examination of lifting equipment: <https://www.hse.gov.uk/work-equipment-machinery/thorough-examinations-lifting-equipment.htm>
- Prolyte truss user manual, Part 1: https://www.prolyte.com/getattachment/Support/Manuals/Download-Prolyte_Trusses_User_manual_Part_1_issue2_.pdf.aspx?lang=en-US